

Samuel Kiyoshi Nishizono-Miller

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Personal Profile

Highly motivated MEng Aerospace graduate with a deep passion for the limitless possibilities of engineering. A proven track record of being a good team player with experience in leadership and public speaking. Eager to start using problem-solving and critical thinking abilities in challenging roles in the engineering industry.

Education

MEng Aerospace Engineering | University of Bristol | Sep 2020 – Jun 2024

- Achieved **1st class with honours**
- Advanced knowledge of aerodynamics including experimental and numerical methods
- Advanced knowledge of composites, structures and materials
- Advanced knowledge of space systems
- Lab Experience in experimental modal analysis and experimental aerodynamics.
- Expert in multiple programming languages and modelling tools (MATLAB, Python, C++, Java, JavaScript, Simulink)
- Skilled experience in CAD software (Inventor, Fusion 360) and FEA software (Abaqus, MSC Patran, MSC Nastran)
- Achieved STK Master Certification as well as skilled experience in GMAT

A Level | Twyford C of E High School | Sep 2018 – Mar 2020

- Physics (A*), Maths (A*), Further Maths (A*), Computer Science (A*) and Japanese (A*)

Projects

Aeolus 2 Design Project | University of Bristol | Sep 2023 – Dec 2023

- Designed the next-generation Aeolus wind profiling weather satellite
- Worked as mission analyst using STK to design the orbit, determine eclipse times and delta-V for orbit maintenance.
- Assisted on the structural analysis (FEA) to determine structural performance

Research on Hemispherical Resonator Gyroscopes (HRG) | University of Bristol | Sep 2022 – May 2023

- Studied the relationship between the change in temperature and the change in resonant frequency of a steel HRG
- Conducted FEA analysis using Abaqus to model change in resonant frequency
- Conducted experimental model analysis in the **Lab** to verify FEA analysis

UKSEDS Olympus Rover Trials | University of Bristol Aether Team | Sep 2021 – Aug 2022

- Developed a Mars Rover for the Mars Sample Collect mission as part of an inter-university competition
- Worked on the communications subsystem and worked as the lead software engineer
- Assisted in the manufacture of the rover and was the operator on the competition day

UAV Wing development and manufacturing | University of Bristol | Sep 2021 – Apr 2022

- Collaborated with 31 students to design the wing for a 2.5m span fixed-wing UAV
- Worked as primary CAD engineer using Fusion 360
- Destructively tested the wing to critical load in the Lab
- Determined wing modes by experimental model analysis

Work Experience

Tutor | Bristol | Sep 2022 - Present

- Taught GCSE and A level for Physics and Mathematics
- Tailored each session to the student and their exam board to successfully improve their grades

Tesco Colleague | Tesco | Golden Hill, Bristol | Jun 2023-May 2024

- Maintained stock availability on shelves
- Assisted customers with inquiries, locating products and offering product advice
- Provided a personalised experience in serving customers at checkout stations

Waiter | Eat Tokyo | Hammersmith, London | Apr 2019-Aug 2019

- Provided excellent customer service to diners delivering food and resolving issues that arose
- Worked as part of the team to keep the restaurant clean during and after business hours
- Prepared takeaway orders for delivery.

Extracurricular Leadership Positions

Japanese Society Secretary | University of Bristol | Sep 23 – Jun 24

- Managed the society's emails. Including the implementation of HTML-formatted emails.
- Took minutes at society meetings.
- Chaired the annual general meeting and ran free and fair elections as well as passing constitutional amendments.
- Successfully increased the membership by 20% to 148 members.

Japanese Society Treasurer | University of Bristol | Sep 22 – Aug 23

- Managed Society funds.
- Successfully kept the society in the black for the first time in 2 years.

Other Activities

Symphonia Symphony Orchestra | University of Bristol | Sep 2022-Jun 2023

- Semi-auditioned student-run orchestra [\[link\]](#).
- Repertoire included Beethoven's 5th, the nutcracker suite, Bruckner's 9th, Mendelssohn's 5th, etc.

Symphonia Philharmonic Orchestra | University of Bristol | Sep 2021-Jun 2023

- Non-auditioned student-run orchestra [\[link\]](#).
- Played as cello section lead for 2 years.

The Maker Challenger | Imperial College Outreach | Aug 2019

- Competition to learn creative and practical skills.
- Placed third place by making a tabletop wind tunnel.
- Learnt a range of skills including but not limited to 3D printing, laser cutting, and basic woodworking.

Volunteer at Oxfam Bookshop | Chiswick, London | Sep 2016 – Mar 2020

- Maintained the sheet music and disc media sections of the store.
- Readied and sorted donated books to be put on the store shelf.

Other Skills

- **Languages:** English (native), Japanese (native), German (entry-level)
- **Other Hard Skills:** Office 365, Git, Data Analysis, Linux/Unix terminal
- **Soft Skills:** Team working, Problem Solving, Work ethic.

Notable Achievement

- **Certificates:** STK Master Certificate
- **Duke of Edinburgh's Award:** Bronze and Silver
- **ABRSM exams:** Piano Grade 8, Cello Grade 7 and Music Theory Grade 5

Hobbies

- **Music:** Cello and Piano
- **Boardgames:** Diplomacy, Chess, etc.
- **Film Photography**
- **Sports:** Cycling and Table Tennis